

Waheed Davon Williams, Ed.D.

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PROFESSIONAL PROFILE

Interdisciplinary scholar practitioner with expertise in engineering, computer science, mathematics, and educational leadership. Experienced in research, data science, and STEM education across K–16 contexts. Passionate about advancing equity and innovation in STEM through culturally responsive pedagogy, data-driven research, and the development of AI-powered learning systems. Skilled at bridging research and practice to support student success, workforce preparation, and cutting-edge computational research.

EDUCATION

Ph.D. (in progress), Engineering & Computational Sciences – Tennessee State University, Expected 2025

M.S., Computer Science – Tennessee State University, Dec 2024

Ed.D., Educational Leadership – Tennessee State University, Dec 2020

Ed.S., Administration & Supervision – Tennessee State University, May 2011

M.S., Mathematical Sciences – Tennessee State University, Aug 2005

B.S., Physics – Tennessee State University, Aug 2001

A.A., Pre-Engineering – College of the Bahamas, May 1997

RESEARCH INTERESTS

STEM Education Research: Culturally responsive teaching, student success in underrepresented populations, AI-powered learning systems, mathematics education at HBCUs.

Computational Sciences: Machine learning, data visualization, relational databases, applied mathematics.

Interdisciplinary Research: generative AI for STEM, structural modeling.

SELECTED RESEARCH & PROJECTS

Machine Learning – Fashion MNIST CNN (2024)

- Implemented and optimized convolutional neural networks (CNNs) with TensorFlow/Keras.
- Conducted model evaluation, hyperparameter tuning, and comparative analysis for performance gains.

World Happiness Report Analysis (2023)

- Applied PCA and data visualization methods to analyze global well-being trends (2015–2019).
- Produced reproducible Python/Jupyter workflows, integrating pandas, seaborn, and matplotlib.

Database Design – Freight Forwarders System (2023)

- Built relational and hybrid database schema in MySQL Workbench.
- Streamlined processes for cargo tracking and logistics operations.

STEM Education Research – Dissertation (2020)

- Investigated perceptions of African American students' mathematics success at an HBCU.
- Contributed to the literature on equity in STEM education and student-centered instructional practices.

Publications & Presentations

- “Travelling Waves of the Perturbed Nonlinear Schrödinger’s Equation,” Intl. Journal of Differential Equations and Applications (2004).
- “Use of X-ray Optics to Determine A-structural Changes of Liquid Crystals,” TSU Research Symposium (2002).

PROFESSIONAL EXPERIENCE

Graduate Assistant, Title III / All Star Math Institute (ASMI) – TSU (2023–2025)

- Provide mathematics instruction and mentoring to diverse program participants.
- Support curriculum development, student recruitment, and assessment of STEM learning outcomes.

Graduate Assistant, Computer Science Department – TSU (2023)

- Assisted faculty with Java programming instruction and grading.

Assistant Professor, Mathematics Education – TSU (2021–2022)

- Developed and delivered undergraduate Calculus and Pre-Calculus courses.
- Designed online learning modules to enhance accessibility and retention.

Instructor, Verizon, STEM Achievers – TSU (Summer 2021, 2023, 2024, 2026)

- Operated and programmed autonomous RVR and Drones hardware to process sensor inputs and navigate complex real-world paths with middle school students.
- Leveraged block-based and text-based python coding language to program and control vehicles with desktop computers and iPads.

Instructor, TRIO Upward Bound – TSU (Summer 2021, Summer 2026)

- Delivered summer instruction in mathematics, physics, and Python programming.
- Prepared underrepresented students for ACT/SAT STEM content.

Adjunct Instructor, Mathematics – TSU (2015–2020)

- Taught Algebra, Pre-Calculus, and Contemporary Math.

Unit Academic Advisor, TRIO Educational Talent Search – TSU (2006–2013, Fall 2025 – Spring 2026)

- Provided academic advising, career counseling, and grant writing support.
- Led math/physics instruction, ACT/SAT prep, and STEM outreach.

Research Assistant, College of Engineering & Technology – TSU (2001–2002)

- Contributed to funded engineering research, created technical flowcharts, and designed web tools.

SKILLS

Technical: Python, TensorFlow/Keras, MATLAB, MySQL, SPSS, Anaconda, Jupyter, Qualtrics, Linux/Windows

Research: Experimental design, statistical analysis, machine learning, database development, qualitative/quantitative methods

STEM Education: Curriculum design, culturally responsive pedagogy, tutoring/mentoring, ACT/SAT prep, online learning platforms

Professional: Leadership, academic advising, program evaluation, grant writing, communication & presentation

AWARDS & HONORS

- ACUE Certificate in Effective College Instruction (2022)
- TRIO Student Support Counselor Awards (2003, 2014)
- Graduate Assistant Scholarships (2003, 2013, 2023)
- Bahamas Electricity Scholarship Recipient (1993)

AFFILIATIONS

- National Society of Black Physicists (NSBP)
- American Mathematical Society (AMS)
- Tennessee Academy of Science
- Verizon Innovative Learning (VIL) Instructor
- Former SMART Evaluator – National Defense Education Program

CERTIFICATES

- The Recreational UAS Safety Test (TRUST) 2026
- Mathematics of AI/ML TSU 2025
- Privacy-Preserving AI 2026
- Mandatory Child Abuse Reporter Training 2026
- Understanding Title VI 2026